

DATABASE PROJECT REPORT

ON-DEMAND STREAMING PLATFORM (MINI-NETFLIX)

Due Date: [29/08 - Week 4]

Project ID & Title: On-Demand Streaming Platform (Mini-Netflix)

Team Name: [ba chang linh ngu lam]

Team Members:

1. [Lê Ngọc Dũng (n24dece065@student.ptithcm.edu.vn)]
2. [Lê Phạm Khánh Huy(n24dece072@student.ptithcm.edu.vn)]
3. [Phan Giảng Bình(n24dece056@student.ptithcm.edu.vn)]

Project Title:

Database Design and Implementation for an On-Demand Streaming Platform

TABLE OF CONTENTS

A. Project Identity

1. Introduction & Project Scope

1.1 System Objective

1.2 Business Rules & Constraints

2. Database Design

2.1 Conceptual Model (ER/EER Diagram)

2.2 Logical Schema Mapping

2.3 Normalization Verification (1NF $\rightarrow$ 3NF/BCNF)

3. Data Dictionary

4. Database Implementation

4.1 DDL Script (Tables, Views, Indexes, Triggers)

4.2 Advanced Queries & Performance Test Cases

5. Verification & Security

5.1 Test Cases

5.2 Role-Based Access Control (RBAC)

A. PROJECT IDENTITY

Item	Information
Project ID & Title	[Insert Project ID]: On-Demand Streaming Platform (Mini-Netflix)
Team Name	Ba chang linh ngu lam
Team Members	Lê Ngọc Dũng – n24dece065@student.ptithcm.edu.vn Phan Giảng Bình – n24dece056@student.ptithcm.edu.vn Lê Phạm Khánh Huy – n24dece072@student.ptithcm.edu.vn
Project Title	Database Design and Implementation for an On-Demand Streaming Platform
Project Domain	Online video streaming / digital entertainment
Database Focus	EER hierarchy, M:N mapping, normalization, constraints, SQL security

1. INTRODUCTION & PROJECT SCOPE

1.1 System Objective

The objective of this project is to design and implement a relational database for an on-demand streaming platform similar to a simplified Netflix system. The database manages user accounts, multiple profiles, content categorization, movies, TV series, episodes, viewing history, subscription plans, and user subscriptions.

The project applies an Enhanced Entity Relationship (EER) model to represent inheritance in the content domain. CONTENT is modeled as a supertype, while MOVIES and EPISODE are subtypes. The many-to-many relationship between PROFILE and CONTENT is implemented through WATCH_HISTORY, which stores viewing progress and completion status.

1.2 Business Rules & Constraints

- Definition:
 1. User: The registered accounts of people who use the streaming platform.
 - a. USER_ID: Unique identifier for a user.
 - b. USERNAME: Login/display username.
 - c. PASSWORD: User's password credential.
 - d. EMAIL: User's email address.
 2. Profile: An individual viewing profile created under a user account.
 - a. PROFILE_ID: Unique profile identifier.
 - b. USER_ID: Owner of the profile.
 - c. PROFILE_NAME: Name of the profile.
 - d. BIRTHDATE: Date of birth of profile user.
 3. Subscription: A user's subscription to the streaming service during a specific period.
 - a. SUBSCRIPTION_ID: Unique subscription identifier.
 - b. USER_ID: User who owns subscription.
 - c. PLAN_ID: Selected subscription plan.

- d. STATUS: Current subscription status.
 - e. START_DATE: Subscription starting date.
 - f. END_DATE: Subscription ending date.
- 4. Subscription plan: Different subscription packages offered by the streaming platform.
 - a. PLAN_ID: Unique plan identifier.
 - b. PLAN_NAME: Name of plan.
 - c. PRICE: Price of subscription.
 - d. QUALITY: Supported video quality.
 - e. MAX_PROFILES: Maximum profiles allowed.
- 5. Watch history: Recording the viewing activities of profiles.
 - a. WATCH_HISTORY_ID: Unique viewing record.
 - b. PROFILE_ID: Profile that watched.
 - c. CONTENT_ID: Content watched.
 - d. WATCH_DURATION: Amount of time watched.
 - e. COMPLETED: Whether viewing was completed.
- 6. Content: The supertype that represents all video content available on the streaming platform.
 - a. CONTENT_ID: Unique content identifier.
 - b. TITLE: Content title.
 - c. DESCRIPTION: Description/synopsis.
 - d. RELEASE_DATE: Original release date.
 - e. DURATION: Duration in minutes.
 - f. CONTENT_TYPE: Type of content.
- 7. Movies: The movie subtype of CONTENT.
 - a. CONTENT_ID: Identifies the movie.
- 8. TV series: A television series available on the streaming platform.
 - a. SERIES_ID: Unique series identifier.
 - b. SERIES_NAME: Name of the series.
- 9. Episode: An individual episode of a TV series.
 - a. CONTENT_ID: Episode's Content ID.
 - b. SERIES_ID: TV Series containing episodes.
 - c. EPISODE_NUMBER: Episode number.
 - d. SEASON_NUMBER: Season number.
- Rules:
 - 1. Users and Profiles:
 - a. A user can create one or more Profiles.
 - b. Each Profile belongs to exactly one User.
 - c. A User may have multiple Profiles, but a Profile cannot belong to multiple Users.
 - d. The number of profiles a User creates should not exceed the MAX_PROFILES allowed by their current subscription plan.
 - 2. Users and Subscriptions:
 - a. A **User** can have one or more subscription records over time.
 - b. Each **Subscription** belongs to exactly one User.

- c. A Subscription uses exactly one **Subscription Plan**.
 - d. A User's subscription has a status such as ACTIVE, EXPIRED, or CANCELLED.
 - e. START_DATE must occur before END_DATE.
 - f. A User should have at most one **active subscription** at a given time.
- 3. Subscription and Subscription Plan:
 - a. Each Subscription must use exactly one Subscription Plan.
 - b. One Subscription Plan can be used by many Subscriptions.
 - c. The price, quality, and maximum number of profiles are determined by the selected plan.
 - d. PRICE must be greater than or equal to 0.
 - e. MAX_PROFILES must be a positive integer.
- 4. Profiles and Watch History:
 - a. A Profile can have many Watch History records.
 - b. Each Watch History record belongs to exactly one Profile.
 - c. A Profile can watch the same Content multiple times if the system allows repeated viewing.
 - d. WATCH_DURATION records how much of the content the profile has watched.
 - e. WATCH_DURATION cannot be negative.
 - f. COMPLETED indicates whether the profile has finished watching the content.
- 5. Watch History and Content:
 - a. Each Watch History record refers to exactly one Content item.
 - b. One Content item can appear in many Watch History records.
 - c. Multiple profiles can watch the same Content.
 - d. A watch history record cannot exist for a Content item that does not exist.
- 6. Content Types:
 - a. Every Movie is a type of Content.
 - b. Every Episode is a type of Content.
 - c. A Movie does not have episode-specific attributes such as EPISODE_NUMBER.
 - d. An Episode has information about its TV Series.
 - e. CONTENT_ID in MOVIES and EPISODE is also a foreign key referencing CONTENT.CONTENT_ID.
- 7. Episodes belong to TV Series:
 - a. Each Episode belongs to exactly one TV Series.
 - b. A TV Series can contain many Episodes.
 - c. An Episode cannot belong to multiple TV Series.
 - d. SERIES_ID in EPISODE is a foreign key referencing TV_SERIES.SERIES_ID.
 - e. EPISODE_NUMBER must be positive.
 - f. SEASON_NUMBER must be positive.

2. DATABASE DESIGN (ISO/IEC 19505 / IE STANDARDS)

2.1 Conceptual Model (ER/EER Diagram)

The conceptual model contains nine main entities: USERS, PROFILE, CONTENT, MOVIES, TV_SERIES, EPISODE, WATCH_HISTORY, SUBSCRIPTION_PLAN, and SUBSCRIPTION. The central EER hierarchy is $\text{CONTENT} \rightarrow \{\text{MOVIES}, \text{EPISODE}\}$.

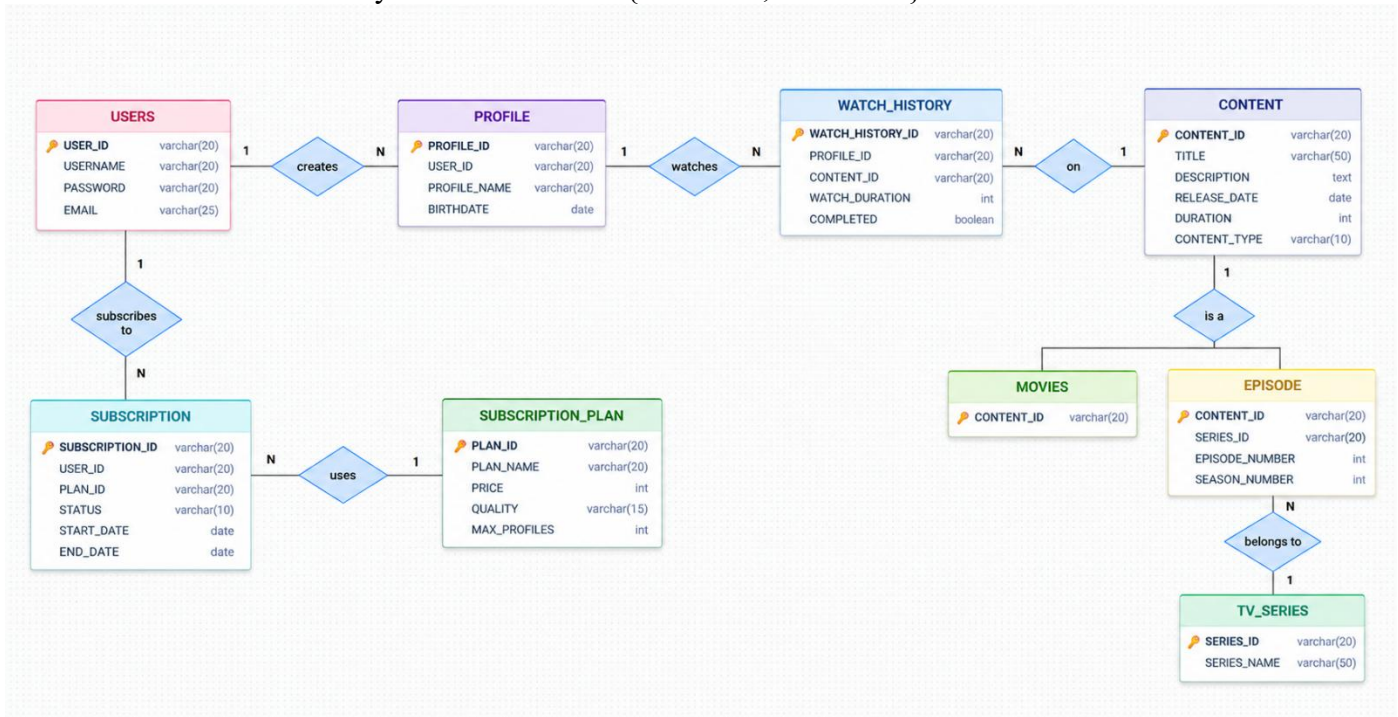


Figure 1. EER Diagram of the Mini-Netflix Database

2.2 Logical Schema Mapping

Relation	Primary Key	Foreign Keys
USERS(user_id, username, password_hash, email)	user_id	—
PROFILE(profile_id, user_id, profile name, birthdate)	profile_id	user_id → USERS
CONTENT(content_id, title, description, release_date, duration)	content_id	—
MOVIES(content_id)	content_id	content_id → CONTENT
TV_SERIES(series_id, series_name)	series_id	—
EPISODE(content_id, series_id, episode number, season number)	content_id	content_id → CONTENT; series_id → TV SERIES
WATCH_HISTORY(watch_history_id, profile_id, content_id, watch_duration, completed)	watch_history_id	profile_id → PROFILE; content_id → CONTENT
SUBSCRIPTION_PLAN(plan_id, plan_name, price, quality, max_profiles)	plan_id	—
SUBSCRIPTION(subscription_id, user_id, plan_id, status, start_date, end_date)	subscription_id	user_id → USERS; plan_id

user_id, plan_id, status, start_date, end_date)		→ SUBSCRIPTION_PLAN
---	--	---------------------

2.3 Normalization Verification (Formal Proofs for 1NF → 3NF/BCNF)

2.3.1 First Normal Form (1NF)

All relations contain atomic values and no repeating groups. Episodes are stored as rows instead of repeated episode columns. Therefore, all relations satisfy 1NF.

2.3.2 Second Normal Form (2NF)

Every relation is in 1NF and all non-key attributes depend on the whole candidate key. Most relations have a single-attribute primary key, so partial dependency does not occur. Therefore, the schema satisfies 2NF.

2.3.3 Third Normal Form (3NF)

Transitive dependencies are removed by separating independent entities. For example, PLAN_NAME, PRICE, QUALITY, and MAX_PROFILES are stored in SUBSCRIPTION_PLAN rather than SUBSCRIPTION. SERIES_NAME is stored in TV_SERIES rather than EPISODE.

USERS:

user_id → username, password_hash, email

PROFILE:

profile_id → user_id, profile_name, birthdate

CONTENT:

content_id → title, description, release_date, duration

TV_SERIES:

series_id → series_name

SUBSCRIPTION_PLAN:

plan_id → plan_name, price, quality, max_profiles

SUBSCRIPTION:

subscription_id → user_id, plan_id, status, start_date, end_date

Each determinant above is a key for its relation and descriptive attributes are stored with the entity they describe. Thus the main schema satisfies 3NF.

2.3.4 BCNF

Under the stated functional dependencies, the determinant of every non-trivial dependency is a candidate key. Therefore, the main relations are compatible with BCNF. Unique constraints on USERNAME, EMAIL, PLAN_NAME, episode position, and profile-content history further improve integrity.

3. DATA DICTIONARY (ADAPTED FROM ISO/IEC 11179)

Table	Attribute	Data Type	Constraints	Definition
USERS	USER_ID	VARCHAR(20)	PK	Unique user identifier
USERS	USERNAME	VARCHAR(20)	UNIQUE, NOT NULL	Login name
USERS	PASSWORD_HASH	VARCHAR(255)	NOT NULL	Password hash
USERS	EMAIL	VARCHAR(100)	UNIQUE, NOT NULL	User email
PROFILE	PROFILE_ID	VARCHAR(20)	PK	Profile identifier
PROFILE	USER_ID	VARCHAR(20)	FK	Owner user
PROFILE	PROFILE_NAME	VARCHAR(20)	NOT NULL	Display name
PROFILE	BIRTHDATE	DATE	NULL	Birth date
CONTENT	CONTENT_ID	VARCHAR(20)	PK	Content identifier
CONTENT	TITLE	VARCHAR(50)	NOT NULL	Content title
CONTENT	DESCRIPTION	TEXT	NULL	Content description
CONTENT	RELEASE_DATE	DATE	NULL	Release date
CONTENT	DURATION	INTEGER	CHECK > = 0	Duration in minutes
MOVIES	CONTENT_ID	VARCHAR(20)	PK, FK	Movie identifier
TV_SERIES	SERIES_ID	VARCHAR(20)	PK	Series identifier
TV_SERIES	SERIES_NAME	VARCHAR(50)	NOT NULL	Series name
EPISODE	CONTENT_ID	VARCHAR(20)	PK, FK	Episode content
EPISODE	SERIES_ID	VARCHAR(20)	FK	Parent series
EPISODE	EPISODE_NUMBER	INTEGER	CHECK > 0	Episode sequence
EPISODE	SEASON_NUMBER	INTEGER	CHECK > 0	Season sequence
WATCH_HISTORY	WATCH_HISTORY_ID	VARCHAR(20)	PK	History identifier
WATCH_HISTORY	PROFILE_ID	VARCHAR(20)	FK	Viewer profile
WATCH_HISTORY	CONTENT_ID	VARCHAR(20)	FK	Viewed

				content
WATCH_HISTORY	WATCH_DURATION	INTEGER	CHECK > = 0	Minutes watched
WATCH_HISTORY	COMPLETED	BOOLEAN	NOT NULL	Completion status
SUBSCRIPTION_PLAN	PLAN_ID	VARCHAR(20)	PK	Plan identifier
SUBSCRIPTION_PLAN	PLAN_NAME	VARCHAR(20)	UNIQUE	Plan name
SUBSCRIPTION_PLAN	PRICE	INTEGER	CHECK > = 0	Plan price
SUBSCRIPTION_PLAN	QUALITY	VARCHAR(15)	NOT NULL	Streaming quality
SUBSCRIPTION_PLAN	MAX_PROFILES	INTEGER	CHECK > 0	Maximum profiles
SUBSCRIPTION	SUBSCRIPTION_ID	VARCHAR(20)	PK	Subscription identifier
SUBSCRIPTION	USER_ID	VARCHAR(20)	FK	Subscribed user
SUBSCRIPTION	PLAN_ID	VARCHAR(20)	FK	Selected plan
SUBSCRIPTION	STATUS	VARCHAR(10)	CHECK	Subscription state
SUBSCRIPTION	START_DATE	DATE	NOT NULL	Start date
SUBSCRIPTION	END_DATE	DATE	CHECK	End date

4. DATABASE IMPLEMENTATION (SQL STYLE GUIDE COMPLIANT)

The following implementation uses PostgreSQL-style SQL. SQL keywords are uppercase and identifiers use snake_case.

4.1 DDL Script (Tables, Views, Indexes, Triggers)

4.1.1 Tables

```
CREATE TABLE users (
    user_id VARCHAR(20) PRIMARY KEY,
    username VARCHAR(20) NOT NULL UNIQUE,
    password_hash VARCHAR(255) NOT NULL,
    email VARCHAR(100) NOT NULL UNIQUE
);

CREATE TABLE subscription_plan (
    plan_id VARCHAR(20) PRIMARY KEY,
    plan_name VARCHAR(20) NOT NULL UNIQUE,
    price INTEGER NOT NULL CHECK (price >= 0),
    quality VARCHAR(15) NOT NULL,
```

```
    max_profiles INTEGER NOT NULL CHECK (max_profiles > 0)
);
```

```
CREATE TABLE profile (
    profile_id VARCHAR(20) PRIMARY KEY,
    user_id VARCHAR(20) NOT NULL,
    profile_name VARCHAR(20) NOT NULL,
    birthdate DATE,
    FOREIGN KEY (user_id) REFERENCES users(user_id) ON DELETE CASCADE
);
```

```
CREATE TABLE content (
    content_id VARCHAR(20) PRIMARY KEY,
    title VARCHAR(50) NOT NULL,
    description TEXT,
    release_date DATE,
    duration INTEGER NOT NULL CHECK (duration >= 0)
);
```

```
CREATE TABLE movies (
    content_id VARCHAR(20) PRIMARY KEY,
    FOREIGN KEY (content_id) REFERENCES content(content_id) ON DELETE CASCADE
);
```

```
CREATE TABLE tv_series (
    series_id VARCHAR(20) PRIMARY KEY,
    series_name VARCHAR(50) NOT NULL
);
```

```
CREATE TABLE episode (
    content_id VARCHAR(20) PRIMARY KEY,
    series_id VARCHAR(20) NOT NULL,
    episode_number INTEGER NOT NULL CHECK (episode_number > 0),
    season_number INTEGER NOT NULL CHECK (season_number > 0),
    FOREIGN KEY (content_id) REFERENCES content(content_id) ON DELETE CASCADE,
    FOREIGN KEY (series_id) REFERENCES tv_series(series_id) ON DELETE CASCADE,
    UNIQUE (series_id, season_number, episode_number)
);
```

```
CREATE TABLE watch_history (
    watch_history_id VARCHAR(20) PRIMARY KEY,
    profile_id VARCHAR(20) NOT NULL,
    content_id VARCHAR(20) NOT NULL,
    watch_duration INTEGER NOT NULL CHECK (watch_duration >= 0),
    completed BOOLEAN NOT NULL DEFAULT FALSE,
    FOREIGN KEY (profile_id) REFERENCES profile(profile_id) ON DELETE CASCADE,
    FOREIGN KEY (content_id) REFERENCES content(content_id) ON DELETE CASCADE,
```

```

        UNIQUE (profile_id, content_id)
    );

CREATE TABLE subscription (
    subscription_id VARCHAR(20) PRIMARY KEY,
    user_id VARCHAR(20) NOT NULL,
    plan_id VARCHAR(20) NOT NULL,
    status VARCHAR(10) NOT NULL
        CHECK (status IN ('ACTIVE', 'EXPIRED', 'CANCELLED')),
    start_date DATE NOT NULL,
    end_date DATE NOT NULL CHECK (end_date >= start_date),
    FOREIGN KEY (user_id) REFERENCES users(user_id),
    FOREIGN KEY (plan_id) REFERENCES subscription_plan(plan_id)
);

```

4.1.2 View

```

CREATE VIEW v_profile_watch_history AS
SELECT p.profile_id, p.profile_name, c.content_id, c.title,
       w.watch_duration, w.completed
FROM watch_history w
JOIN profile p ON p.profile_id = w.profile_id
JOIN content c ON c.content_id = w.content_id;

```

4.1.3 Indexes

```

CREATE INDEX idx_profile_user ON profile(user_id);
CREATE INDEX idx_episode_series ON episode(series_id, season_number, episode_number);
CREATE INDEX idx_watch_history_profile ON watch_history(profile_id);
CREATE INDEX idx_watch_history_content ON watch_history(content_id);
CREATE INDEX idx_subscription_user_status ON subscription(user_id, status);

```

4.1.4 Trigger

```

CREATE OR REPLACE FUNCTION validate_watch_duration()
RETURNS TRIGGER AS $$
DECLARE max_duration INTEGER;
BEGIN
    SELECT duration INTO max_duration
    FROM content
    WHERE content_id = NEW.content_id;

    IF NEW.watch_duration > max_duration THEN
        RAISE EXCEPTION 'Watch duration cannot exceed content duration';
    END IF;
    RETURN NEW;
END;
$$ LANGUAGE plpgsql;

CREATE TRIGGER trg_validate_watch_duration
BEFORE INSERT OR UPDATE ON watch_history

```

```
FOR EACH ROW EXECUTE FUNCTION validate_watch_duration();
```

4.2 Advanced Queries & Performance Test Cases

4.2.1 Viewing History of a Profile

```
SELECT p.profile_name, c.title, w.watch_duration, w.completed
FROM watch_history w
JOIN profile p ON p.profile_id = w.profile_id
JOIN content c ON c.content_id = w.content_id
WHERE p.profile_id = 'P001';
```

4.2.2 Episodes of a TV Series

```
SELECT ts.series_name, e.season_number, e.episode_number, c.title
FROM episode e
JOIN tv_series ts ON ts.series_id = e.series_id
JOIN content c ON c.content_id = e.content_id
WHERE e.series_id = 'S001'
ORDER BY e.season_number, e.episode_number;
```

4.2.3 Active Subscription for Each User

```
SELECT u.username, sp.plan_name, sp.quality, s.start_date, s.end_date
FROM subscription s
JOIN users u ON u.user_id = s.user_id
JOIN subscription_plan sp ON sp.plan_id = s.plan_id
WHERE s.status = 'ACTIVE';
```

4.2.4 Most Watched Content

```
SELECT c.content_id, c.title, COUNT(*) AS viewer_count,
       SUM(w.watch_duration) AS total_watch_duration
FROM watch_history w
JOIN content c ON c.content_id = w.content_id
GROUP BY c.content_id, c.title
ORDER BY viewer_count DESC, total_watch_duration DESC
LIMIT 10;
```

4.2.5 Performance Test Cases

Compare query execution before and after indexes using EXPLAIN ANALYZE. Actual measurements should be captured from the team's database environment.

```
EXPLAIN ANALYZE
SELECT * FROM watch_history WHERE profile_id = 'P001';
```

```
EXPLAIN ANALYZE
SELECT * FROM episode
WHERE series_id = 'S001'
ORDER BY season_number, episode_number;
```

5. VERIFICATION & SECURITY

5.1 Test Cases Showing Constraints Prevent Bad Data Insertion

Test ID	Invalid Input	Expected Result	Constraint
TC01	Duplicate USERNAME	Rejected	UNIQUE(username)
TC02	Duplicate EMAIL	Rejected	UNIQUE(email)
TC03	Unknown USER_ID in PROFILE	Rejected	Foreign key
TC04	EPISODE_NUMBER = 0	Rejected	CHECK > 0
TC05	Negative PRICE	Rejected	CHECK price >= 0
TC06	END_DATE before START_DATE	Rejected	CHECK dates
TC07	Invalid STATUS	Rejected	CHECK status
TC08	Negative WATCH_DURATION	Rejected	CHECK >= 0
TC09	WATCH_DURATION > CONTENT duration	Rejected	Trigger
TC10	Duplicate episode position	Rejected	UNIQUE(series, season, episode)

5.2 Role-Based Access Control (RBAC) Definition

The RBAC model separates administration, content management, and ordinary application usage.

```
CREATE ROLE db_admin;  
CREATE ROLE content_manager;  
CREATE ROLE application_user;
```

```
GRANT ALL PRIVILEGES  
ON ALL TABLES IN SCHEMA public  
TO db_admin;
```

```
GRANT SELECT, INSERT, UPDATE  
ON content, movies, tv_series, episode  
TO content_manager;
```

```
GRANT SELECT  
ON subscription_plan  
TO content_manager;
```

```
GRANT SELECT  
ON content, movies, tv_series, episode, subscription_plan  
TO application_user;
```

```
GRANT SELECT, INSERT, UPDATE  
ON profile, watch_history
```

```
TO application_user;
```

```
REVOKE SELECT (password_hash)
```

```
ON users
```

```
FROM application_user;
```

In a production environment, password hashes must never be exposed to ordinary users. Direct table access should preferably be replaced by secure APIs, views, or stored procedures.

CONCLUSION

This report presents a complete database project design for an On-Demand Streaming Platform (Mini-Netflix). The main EER feature is the CONTENT supertype with MOVIES and EPISODE as subtypes. The many-to-many relationship between PROFILE and CONTENT is implemented through WATCH_HISTORY.

The schema is normalized through 3NF and is compatible with BCNF under the stated functional dependencies. The implementation includes DDL, constraints, indexes, a view, a trigger, advanced queries, negative test cases, and RBAC definitions. The design therefore follows the required report structure and provides a strong foundation for implementation and future extension.